Sustainability Economics: Complete Lecture Notes

John Janmaat
Economics, Philosophy and Political Science, The University of British Columbia

Academic Year 2026–2027

Contents

Course Overview & Syllabus Summary **v**

I Foundations & Basic Principles **1**

1 Introduction & Key Concepts **3**
1.1 Overview & Objectives 3
1.1.1 Detailed Explanation of Concept 1 4

2 Hermit Economics **5**
2.1 Learning Objectives 5
2.2 The Hermit 5
2.2.1 Detailed Explanation of Concept 1 5

Course Overview & Syllabus Summary

What is the connection between economics and sustainability? When people talk about economics, often they are talking about something to do with money. How people acquire it and what they do with it. When people talk about sustainability, often they are talking about the environment, how it is changing, and whether or not those changes are good. Where do these two things come together? That intersection often seems to be about the way that people are acquiring and using money is changing the environment in ways that are not good.

Part I

Foundations & Basic Principles

Chapter 1

Introduction & Key Concepts

1.1 Overview & Objectives

The term *sustainability* implies continuity and balance. If something is sustainable, it can continue doing what it is doing. Unless it is doing nothing, what it is doing is interacting with other entities. This implies that, except for those things that are doing nothing, we should think about sustainability with a systems perspective.

A system is a collection of entities (or elements) and the connections between them. To understand a system, we need to understand the entities and their connections. Our goal in this course is to develop an understanding of the relationship between the economy and the environment by looking at the elements that make up the economy and their relationship with elements that make up the environment. To do so, we will proceed by building a model of a *complex adaptive social ecological system*.

The system we inhabit, and that we will try to understand, is complex. This means that the system as a whole exhibits properties that cannot be explained by looking at any element itself. It is adaptive in that it changes in response to internal and external pressures.

ey reading for this section: Chapter 1 of the required textbook.

Systems

- A system is a collection of elements and their connections.
- A social ecological system contains elements in both the social (human) and ecological (nature) domains, and has connections crossing between the domains.
- A system is complex when there are 'emergent' properties that cannot be predicted looking at elements in isolation.
- A system is adaptive if it continuously evolves or changes in response to internal and external forces.

Systems



1.1.1 Detailed Explanation of Concept 1

While the slide highlights the main points, it is important to note that the theoretical model relies on two primary assumptions...

Mathematical Representation

$$f(x) = \alpha + \beta x + \varepsilon$$

Chapter 2

Hermit Economics

2.1 Learning Objectives

- Explain the meaning of opportunity cost.
- Demonstrate an understanding of the production function and the production possibilities frontier.
- Explain the role of diminishing marginal product.

2.2 The Hermit

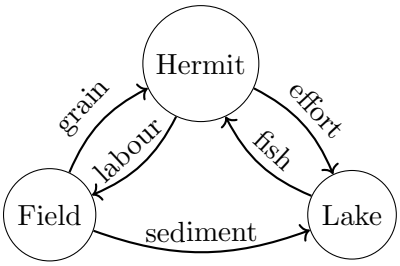
A hermit is someone who has chosen to live apart from society. In the most extreme case, they have no contact with any other person and are completely self sufficient. From a system thinking perspective, the hermit would be one element of the system, with the other elements being part of the ecological side of the hermit-nature social ecological system.

Hermit System

- Hermit,
  - The social element of the social ecological system.
  - Contributes labour and consumes the fruits of that labour.
- Field,
  - One ecological element of the system.
  - Produces grain, with labour as the main input.
- Lake.
  - One ecological element of the system.
  - Produces fish, with fishing effort (time spent fishing) as the main input.

The hermit can apply their time either to labour in the field, growing grain, or effort at the lake to catch fish. The more labour applied in the field, the more grain that the hermit harvests. Likewise, the more effort devoted to fishing, the more fish that will be caught.

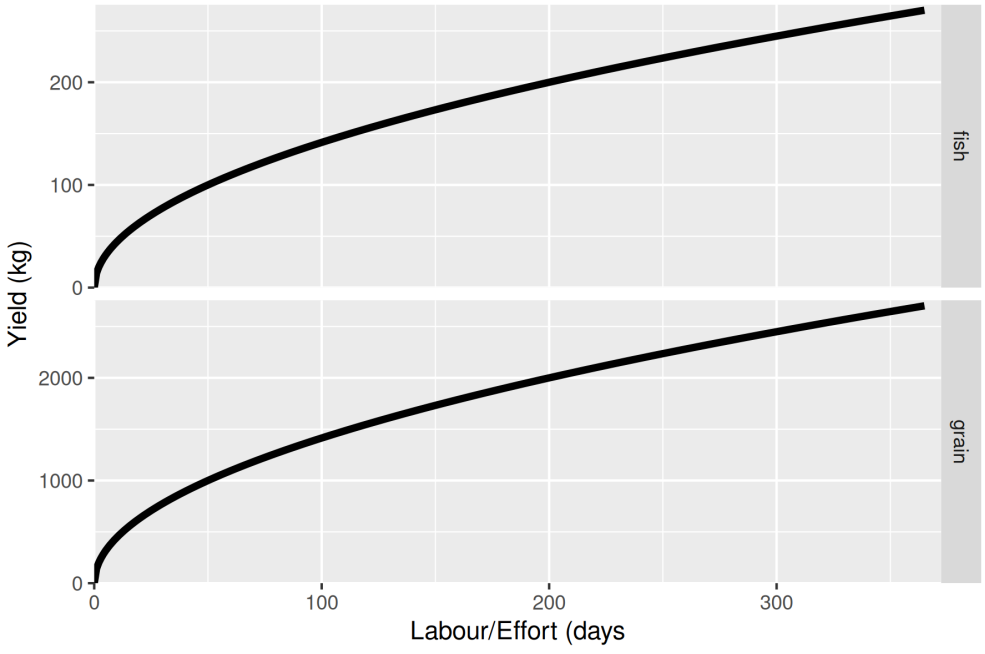
Hermit System



Production Functions

- A *technology* is a way that inputs are converted to outputs.
- A *production function* is a way of representing the technology.
  - The production function is a simplification.
  - The technology behaves the way that physics, chemistry, etc. require it to.
  - All models are wrong, some are useful.
- Three measures are commonly used to describe production at each input level.
  - *Total Product* is the total output produced given a particular input level.
  - *Average Product* is the total product divided by the input level.
  - *Marginal Product* is how much the total output changes if the level of input is changed by a small amount.

Production Functions



Total, Average and Marginal Product

- Total Product (TP) and Average Product (AP) are based on a level of input.

$$AP = TP/L$$

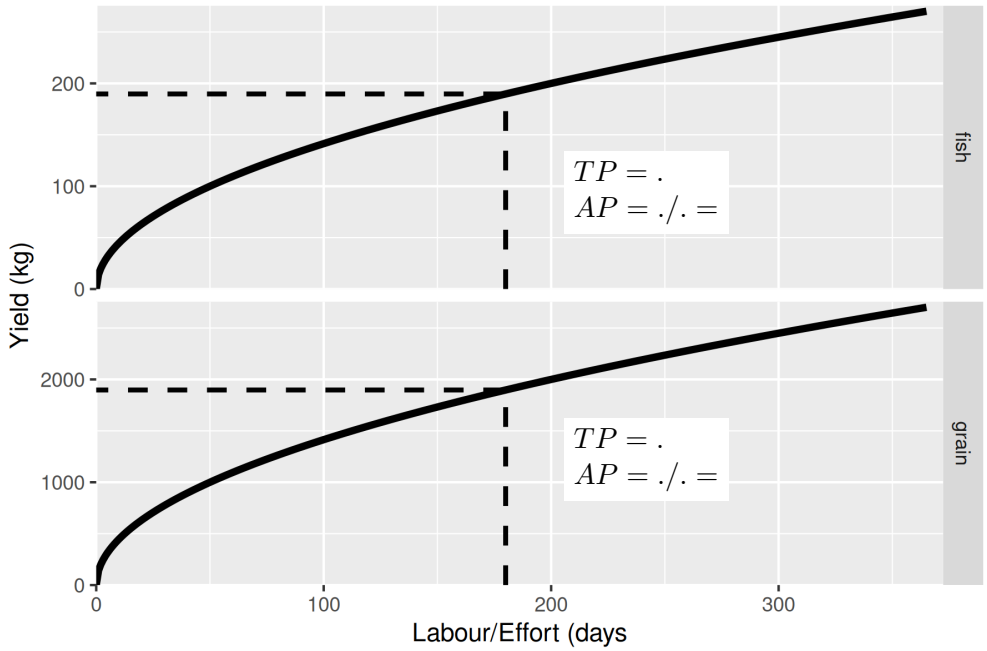
  - L can be any input, not just labour.

- Marginal Product (MP) is based on a small change in in the level of an input, around a specific level of that input.

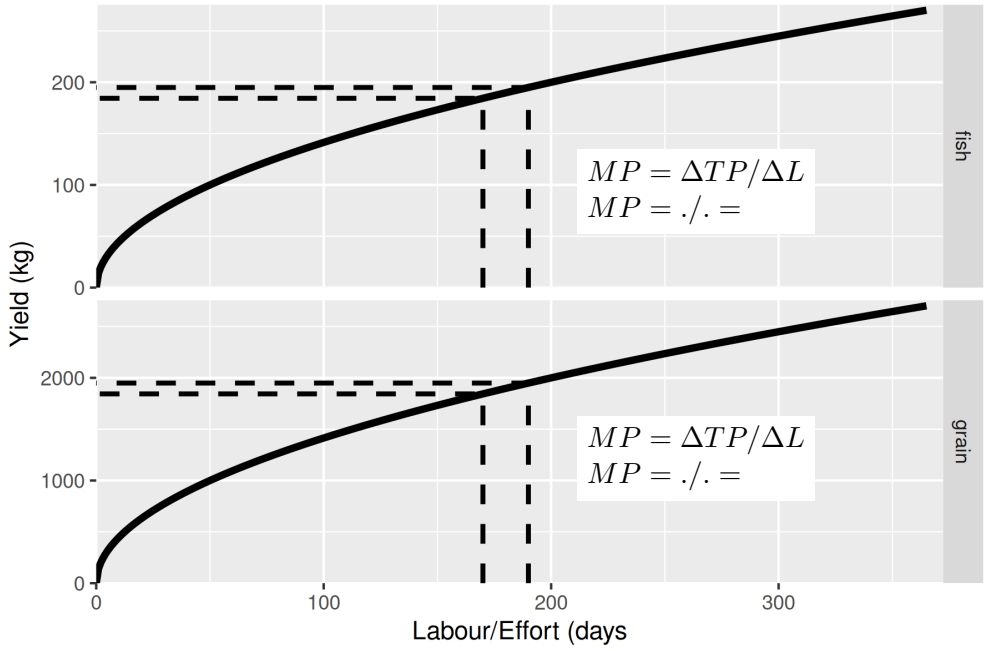
$$MP = \Delta TP/\Delta L = \partial TP/\partial L$$

  - The marginal product measures how much extra output would be produced if a small amount of extra input is used.
  - It is the slope of a line that just touches the production function at the input level.

Total, Average and Marginal Product



Total, Average and Marginal Product



2.2.1 Detailed Explanation of Concept 1

While the slide highlights the main points, it is important to note that the theoretical model relies on two primary assumptions...